

Using the commutation relations for the components of the angular momentum, we have found that the allowed eigenvalues for $\hat{J}^2$ are $\hbar^2 j(j+1)$, where $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$. For each value of j , the eigenvalues of J_z are $\hbar m$, with $m = -j, -j+1, \dots, j-1, j$.

Compared with the solutions of the eigensystem discussed in the [previous chapter](#), we see that we have found more solutions than there are in [Eq. \(8.27\)](#). In this chapter we are going to discuss the reason why the orbital angular momentum can only take integer values and then we are going to see that the idea of angular momentum can be extended in a way that allows half-integer values too. Note that the orbital angular momentum is expressed as a function of the position and momentum operators, i.e. it probes the spatial distribution of the quantum system. When working in position space, [Eq. \(8.27\)](#) is a partial differential equation in θ and ϕ . The solutions of this equation are the spherical harmonics $Y_\ell^m(\theta, \phi)$, [Eq. \(8.26\)](#). The explicit expression for the spherical harmonics shows that the ϕ dependence is simply

$$Y_\ell^m(\theta, \phi) \propto \exp(im\phi), \quad (9.1)$$

as expected, since the spherical harmonics are *also* eigenfunctions of $L_z = -i\hbar \frac{\partial}{\partial \phi}$. Since we required that wave functions must be single-valued, the spherical harmonics must be periodic in ϕ with period 2π :

$$Y_\ell^m(\theta, \phi) = Y_\ell^m(\theta, \phi + 2\pi). \quad (9.2)$$

[Equation \(9.2\)](#) requires m to be an integer, and hence ℓ must also be an integer.

However, as soon as we consider the angular momentum as an intrinsic property of the system, no longer related to the spatial dependence of its wave function, half-integer values of j become possible. This intrinsic angular momentum is called **spin**.¹ An obvious question at this point is: Is there any evidence in nature of systems with non-integer angular momentum?

9.1 The Stern–Gerlach Experiment

This experiment, performed for the first time in 1922 by Stern and Gerlach, provided evidence that the component of the spin along a given direction, S_z , is quantised and its spectrum allows only two values $\pm\hbar/2$, and therefore corresponds to a state with total

¹ When dealing with spin we denote the spin angular momentum vector by $\underline{S}$ and its square by S^2 .

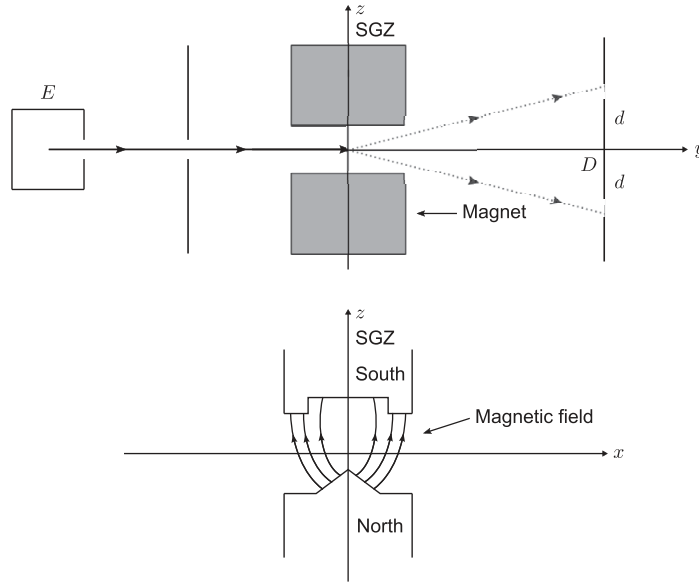


Fig. 9.1

A beam of atoms, each with $j = \frac{1}{2}$, leave the emitter (E) with random spin orientation. They travel along the y -axis passing through a Stern–Gerlach magnet oriented along the z -direction (SGZ). The beam is split by SGZ into only two possible directions, arriving at the detector at D a distance d either above or below the y -axis.

angular momentum $\frac{1}{2}$. It is instructive to discuss the Stern–Gerlach experiment and its interpretation according to the postulates of quantum mechanics.

The Stern–Gerlach experiment is schematically represented in Fig. 9.1. A beam of atoms travels along the y -direction and is deflected by a variable magnetic field $\underline{B}$ in the z -direction. The atoms travel in a potential

$$V = \gamma \underline{S} \cdot \underline{B}, \quad (9.3)$$

where γ , the gyromagnetic ratio, is the proportionality coefficient that relates the magnetic moment of the atom to its spin. The deflection of the atoms, d , either above or below the y -axis, measured by the detector at D is proportional to the force acting on the system, i.e. the gradient of the potential V :

$$d \propto \gamma S_z \frac{\partial B_z}{\partial z}. \quad (9.4)$$

The important point here is that the deflection is proportional to the z -component of the spin of the system, all other factors in Eq. (9.4) being fixed in the experiment. The Stern–Gerlach experiment is a measurement of the z -component of the spin of the atoms.

As usual in quantum mechanics, we prepare the system in the same identical state, perform the experiment multiple times and study the probability distribution of the outcomes of the measurement. The probability distribution for the deflection, and therefore for the value of S_z , is peaked around two values, symmetrical with respect to the y -axis, which we interpret as a quantised spectrum with two possible eigenvalues, $S_z = \pm \hbar/2$, and

therefore the atoms must be systems with spin $S = \frac{1}{2}$. We will come back to the Stern–Gerlach experiment in later chapters.

In order to understand the physical meaning of states with $S = \frac{1}{2}$, we will now investigate their properties in more detail.

9.2 Physical States

As far as the spin degrees of freedom are concerned, the system is a two-state system, with a basis made up of simultaneous eigenvectors of $\hat{S}^2$ and $\hat{S}_z$:

$$\left\{ |S = \frac{1}{2}, S_z = +\frac{1}{2}\rangle, |S = \frac{1}{2}, S_z = -\frac{1}{2}\rangle \right\}. \quad (9.5)$$

Remember that with our conventions, we have

$$\begin{aligned} \hat{S}^2 |S = \frac{1}{2}, S_z = +\frac{1}{2}\rangle &= \frac{3}{4} \hbar^2 |S = \frac{1}{2}, S_z = +\frac{1}{2}\rangle, \\ \hat{S}_z |S = \frac{1}{2}, S_z = +\frac{1}{2}\rangle &= +\frac{1}{2} \hbar |S = \frac{1}{2}, S_z = +\frac{1}{2}\rangle, \\ \hat{S}^2 |S = \frac{1}{2}, S_z = -\frac{1}{2}\rangle &= \frac{3}{4} \hbar^2 |S = \frac{1}{2}, S_z = -\frac{1}{2}\rangle, \\ \hat{S}_z |S = \frac{1}{2}, S_z = -\frac{1}{2}\rangle &= -\frac{1}{2} \hbar |S = \frac{1}{2}, S_z = -\frac{1}{2}\rangle. \end{aligned}$$

It is economical to introduce a shorthand notation for these basis vectors:

$$|S = \frac{1}{2}, S_z = +\frac{1}{2}\rangle = |+\frac{1}{2}\rangle = |\uparrow\rangle, \quad (9.6)$$

$$|S = \frac{1}{2}, S_z = -\frac{1}{2}\rangle = |-\frac{1}{2}\rangle = |\downarrow\rangle. \quad (9.7)$$

A generic state of a spin- $\frac{1}{2}$ system can be written as a linear combination of the basis vectors:

$$|\psi\rangle = \psi_1 |\uparrow\rangle + \psi_2 |\downarrow\rangle, \quad (9.8)$$

where $\psi_1, \psi_2 \in \mathbb{C}$. As usual, having chosen a basis, the vector can be represented by its coordinates:

$$|\psi\rangle \rightarrow \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}, \quad (9.9)$$

and we use the arrow once again to stress the fact that $|\psi\rangle$ is represented by a complex vector *in a given basis*. It is important to emphasise once again that spin is not related to the spatial degrees of freedom of the system. The spin states are vectors in a two-dimensional Hilbert space $\mathcal{H}_{\text{spin}}$, while the states that correspond to the motion of the particle in space are vectors in an infinite-dimensional vector space $\mathcal{H}_{\text{space}}$, which are described for example by a wave function. If we are interested in describing both the spatial distribution and the spin degrees of freedom of the system, then we need to consider states in the tensor product

$$\mathcal{H} = \mathcal{H}_{\text{space}} \otimes \mathcal{H}_{\text{spin}}. \quad (9.10)$$

A basis for $\mathcal{H}$, made up of the tensor product of eigenstates of position and eigenstates of $\hat{S}_z$, is given by

$$\left\{ |\underline{r}, s_z\rangle = |\underline{r}\rangle |s_z\rangle, \underline{r} \in \mathbb{R}^3, s_z = \pm \frac{1}{2} \right\}, \quad (9.11)$$

and a generic state can be expanded as

$$|\Psi\rangle = \int d^3r' \sum_{s_z} |\underline{r}', s_z\rangle \langle \underline{r}', s_z | \Psi \rangle, \quad (9.12)$$

$$= \int d^3r' \left(\psi_1(\underline{r}') |\underline{r}'\rangle |\uparrow\rangle + \psi_2(\underline{r}') |\underline{r}'\rangle |\downarrow\rangle \right), \quad (9.13)$$

where in the second line we have written explicitly the sum over the two values of s_z and used the notation with the arrows introduced above for eigenstates of $\hat{S}_z$. Clearly, in Eq. (9.13)

$$\psi_1(\underline{r}') = \langle \underline{r}', \uparrow | \Psi \rangle, \quad (9.14)$$

$$\psi_2(\underline{r}') = \langle \underline{r}', \downarrow | \Psi \rangle. \quad (9.15)$$

Starting from Eq. (9.13), taking the scalar product with an eigenstate of position and using the orthonormality of position eigenstates yields

$$\langle \underline{r} | \Psi \rangle = \psi_1(\underline{r}) |\uparrow\rangle + \psi_2(\underline{r}) |\downarrow\rangle. \quad (9.16)$$

Once again we can represent $\langle \underline{r} | \Psi \rangle$ as a column vector in the specific basis in $\mathcal{H}_{\text{spin}}$:

$$\langle \underline{r} | \Psi \rangle \longrightarrow \begin{pmatrix} \psi_1(\underline{r}) \\ \psi_2(\underline{r}) \end{pmatrix}. \quad (9.17)$$

The physical interpretation of these components is the standard one, i.e. the coordinates in a given basis are probability amplitudes. Hence

1. $|\psi_1(\underline{r})|^2$ yields the probability density of finding the system at position $\underline{r}$ with spin $s_z = +\frac{1}{2}$.
2. $|\psi_2(\underline{r})|^2$ yields the probability density of finding the system at position $\underline{r}$ with spin $s_z = -\frac{1}{2}$.
3. The probability of finding the system anywhere in space with spin $s_z = +\frac{1}{2}$ is

$$P_{\uparrow} = \int d^3r |\psi_1(\underline{r})|^2. \quad (9.18)$$

4. The probability density of finding the system at position $\underline{r}$ with any value of s_z is

$$P(\underline{r}) = |\psi_1(\underline{r})|^2 + |\psi_2(\underline{r})|^2. \quad (9.19)$$

9.3 Matrix Representation

For $s = \frac{1}{2}$, let us compute the matrix elements $\langle \frac{1}{2}, m' | \hat{S}_i | \frac{1}{2}, m \rangle$; the possible values for m' and m are $m' = \frac{1}{2}$ or $-\frac{1}{2}$ and $m = \frac{1}{2}$ or $-\frac{1}{2}$. If we choose the convention that the row and

column labels start with the largest value of s_z and decrease, so that the first row (column) is labelled $m'(m) = \frac{1}{2}$ and the second row (column) is labelled $m'(m) = -\frac{1}{2}$, we find the following 2×2 matrix representing $\hat{S}_z$:

$$\hat{S}_z \rightarrow \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

We say that this matrix *represents* the operator $\hat{S}_z$ in the $s = \frac{1}{2}$ multiplet.

In order to compute the matrix representations of $\hat{S}_x$ and $\hat{S}_y$, it is easier to start from the raising and lowering operators. The only non-zero element for the matrix representing $\hat{S}_+$ is when $m' = \frac{1}{2}$ and $m = -\frac{1}{2}$, for which

$$c_+ = \sqrt{\frac{1}{2}(\frac{1}{2} + 1) + \frac{1}{2}(\frac{1}{2})} = 1,$$

and hence the matrix is

$$\hat{S}_+ \rightarrow \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

whilst the only non-zero element for the matrix representing $\hat{S}_-$ is when $m' = -\frac{1}{2}$ and $m = \frac{1}{2}$, for which

$$c_- = \sqrt{\frac{1}{2}(\frac{1}{2} + 1) + \frac{1}{2}(\frac{1}{2})} = 1,$$

also giving

$$\hat{S}_- \rightarrow \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

From these two matrices it is easy to construct the matrices representing $\hat{S}_x$ and $\hat{S}_y$, since

$$\begin{aligned} \hat{S}_x &= \frac{1}{2}(\hat{S}_+ + \hat{S}_-), \\ \hat{S}_y &= \frac{1}{2i}(\hat{S}_+ - \hat{S}_-). \end{aligned}$$

Thus

$$\hat{S}_x \rightarrow \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{S}_y \rightarrow \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

You can readily verify that these 2×2 matrices satisfy the angular momentum commutation relations from which we started. We say, therefore, that they *provide a matrix representation* of the angular momentum operators.

The set of three 2×2 matrices which appear above in the matrix representations of $\hat{S}_x$, $\hat{S}_y$ and $\hat{S}_z$ are known as the **Pauli spin matrices** and are usually denoted as follows:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Collectively, we can write

$$\hat{S} \rightarrow \frac{1}{2}\hbar \underline{\sigma},$$

meaning $\hat{S}_x \rightarrow \frac{1}{2}\hbar \sigma_x$, etc. Often we will just write $=$ instead of $\rightarrow$, but you should remember that this is just one possible choice for representing the operators, dictated by the choice of basis we made above in $\mathcal{H}_{\text{spin}}$.

The Pauli matrices are Hermitian, as expected since they are associated with an observable, and have the following property, which you can easily verify:

$$\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = 1,$$

where 1 denotes the unit 2×2 matrix.

9.4 Eigenvectors

We can now rederive the eigenvectors using the matrix representation introduced above. It is trivial to show that the matrix σ_z has eigenvectors which are just two-component column vectors:

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

so that the eigenvalue equations for $\hat{S}_z$ are

$$\frac{1}{2}\hbar \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2}\hbar \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \frac{1}{2}\hbar \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{1}{2}\hbar \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

We see explicitly that $\hat{S}_z$ has eigenvalues $\pm \frac{1}{2}\hbar$, as it should for a system with $s = \frac{1}{2}$, and that the basis vectors are the eigenstates of $\hat{S}_z$.

Furthermore, if we construct the matrix representing $\hat{S}^2$, we see that it has these same two column vectors as eigenvectors with a common eigenvalue $s(s+1)\hbar^2 \equiv \frac{3}{4}\hbar^2$:

$$\hat{S}^2 \equiv \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2 = \frac{1}{4}\hbar^2 \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}^2 + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^2 \right] = \frac{3}{4}\hbar^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and

$$\frac{3}{4}\hbar^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{3}{4}\hbar^2 \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \frac{3}{4}\hbar^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{3}{4}\hbar^2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

We can thus identify the two column vectors with the two simultaneous eigenstates of the operators $\hat{S}^2$ and $\hat{S}_z$:

$$|s = \frac{1}{2}, m = \frac{1}{2}\rangle \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |s = \frac{1}{2}, m = -\frac{1}{2}\rangle \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

As already discussed previously, an arbitrary state $|\psi\rangle$ with $s = \frac{1}{2}$ may be represented as a linear combination of these two states since they span the two-dimensional space of

two-component column vectors:

$$|\psi\rangle = \sum_{m=-\frac{1}{2}}^{\frac{1}{2}} c_m |s = \frac{1}{2}, m\rangle \quad (9.20)$$

is represented by

$$\begin{pmatrix} c_{\frac{1}{2}} \\ c_{-\frac{1}{2}} \end{pmatrix} = c_{\frac{1}{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_{-\frac{1}{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

9.5 Scalar Products

The Dirac kets we have seen are represented by column vectors of rank $(2s + 1)$; the corresponding conjugates, or Dirac bras, are represented by row vectors of the same rank. The rule for scalar products is that if

$$|\psi\rangle \longrightarrow \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}, \quad \text{then} \quad \langle\psi| \longrightarrow (\psi_1^* \quad \psi_2^*).$$

The scalar product of two states $|\psi\rangle$ and $|\phi\rangle$ is then defined to be

$$\langle\phi|\psi\rangle \equiv (\phi_1^* \quad \phi_2^*) \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = \phi_1^* \psi_1 + \phi_2^* \psi_2. \quad (9.21)$$

Thus, for example, for a normalised state

$$\langle\psi|\psi\rangle = \psi_1^* \psi_1 + \psi_2^* \psi_2 = |\psi_1|^2 + |\psi_2|^2 = 1.$$

The orthonormality property of the eigenvectors is also obvious and may be used to project out the coefficients c_m in the expansion of the arbitrary state $|\psi\rangle$ in Eq. (9.20).

9.6 Eigenvectors of $\hat{S}_x$

So far we have concentrated on the eigenvalues and eigenstates of $\hat{S}_z$, but what of the other Cartesian components of angular momentum? It is clear that, since there is nothing special about the z -direction ('space is isotropic!'), we should also expect that measuring say the x -component of the angular momentum for a system with $s = \frac{1}{2}$ can only yield either $\pm \frac{1}{2}\hbar$. Let us verify this. The matrix representing $\hat{S}_x$ is $\frac{1}{2}\hbar \sigma_x$, so we need to find the eigenvalues and eigenvectors of the 2×2 matrix σ_x . Let us write

$$\sigma_x \chi = \rho \chi \quad \text{with} \quad \chi = \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix},$$

where ρ denotes an eigenvalue and χ the corresponding eigenvector. We find the eigenvalues by rewriting this as

$$(\sigma_x - \rho \mathbf{1}) \chi = 0,$$

where $\mathbf{1}$ denotes the unit 2×2 matrix. This is a pair of simultaneous equations for χ_1 and χ_2 , which only have a non-trivial solution if the determinant of the 2×2 coefficient matrix on the left-hand side is singular. The condition for this is

$$\det(\sigma_x - \rho \mathbf{1}) = \begin{vmatrix} -\rho & 1 \\ 1 & -\rho \end{vmatrix} = 0,$$

which yields

$$\rho^2 - 1 = 0 \quad \text{implying that} \quad \rho = \pm 1.$$

The eigenvalues of $\hat{S}_x$ are thus $\pm \frac{1}{2}\hbar$ as anticipated. *More generally, the eigenvalues of $\hat{S}_x$ are written $m_x \hbar$, where we have factored out the dimensionful unit of angular momentum $\hbar$ and m_x is a dimensionless number. In this case, where $s = \frac{1}{2}$, we have $m_x = \pm \frac{1}{2}$. Let us now find the eigenvectors corresponding to the two eigenvalues.*

Case $\rho = 1$

The equation for the eigenvectors becomes

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} = \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} \Rightarrow \begin{pmatrix} \chi_2 \\ \chi_1 \end{pmatrix} = \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} \Rightarrow \chi_2 = \chi_1.$$

We can pick any two-component column matrix which satisfies this condition. In particular, a suitably normalised eigenvector which represents the state with $s = \frac{1}{2}$ and $m_x = \frac{1}{2}$ is

$$|s = \frac{1}{2}, m_x = \frac{1}{2}\rangle \longrightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Case $\rho = -1$

In close analogy with the computation above, let us write

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} = - \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} \Rightarrow \begin{pmatrix} \chi_2 \\ \chi_1 \end{pmatrix} = - \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} \Rightarrow \chi_2 = -\chi_1.$$

Thus a suitably normalised eigenvector which represents the state with $s = \frac{1}{2}$ and $m_x = -\frac{1}{2}$ is

$$|s = \frac{1}{2}, m_x = -\frac{1}{2}\rangle \longrightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Comments

Let us briefly comment on the solutions found above.

- The eigenvectors corresponding to $m_x = \frac{1}{2}$ and $m_x = -\frac{1}{2}$ are orthogonal, as they must be:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0.$$

- The eigenvectors of $\hat{S}_x$ are expressible as linear combinations of the eigenvectors of $\hat{S}_z$:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Thus if a system with $s = \frac{1}{2}$ is in an eigenstate of $\hat{S}_x$, for example with $m_x = \frac{1}{2}$, then the probability that a measurement of $\hat{S}_z$ yields the result $m = \frac{1}{2}$ is $|\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$.

What has emerged from this analysis is that we can consider systems with $s = \frac{1}{2}$ as having *intrinsic* angular momentum, which has nothing to do with the orbital motion of the particle about some point. It is a property of the system in its own rest frame, and can be understood as a new degree of freedom of the particle. Hence the quantum numbers describing the state of the system must also include an index m labelling the values of these degrees of freedom. For the case $s = \frac{1}{2}$, m can take two values, and therefore the wave functions have two components as discussed above. For the general case of spin s , the wave functions have $2s + 1$ components. This intrinsic angular momentum is known as *spin* and does not have any classical analogue. Electrons, protons, neutrons and many of the more unstable particles have spin $\frac{1}{2}$.

9.7 The Stern–Gerlach Experiment Reloaded

We can now revisit the result of the original Stern–Gerlach experiment, which was conducted with a beam of silver atoms and found two emergent beams, corresponding to $s = \frac{1}{2}$. We are going to consider here a more elaborate experiment involving not one but several Stern–Gerlach magnets, which we use to make *successive* measurements of various components of angular momentum.

We assume that we can neglect any interaction between the particles in the beam. The two beams emerging from the first magnet have $m = \frac{1}{2}$ and $m = -\frac{1}{2}$, respectively, but only the former is allowed to proceed to the second magnet. Thus, *we know that each particle entering the second magnet is in the state with $m = \frac{1}{2}$* , represented by the column vector

$$|s = \frac{1}{2}, m = \frac{1}{2}\rangle \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The second magnet, which has its mean field aligned with the x -direction, measures the x -component of angular momentum. We can predict the outcome by expanding the state $|s = \frac{1}{2}, m = \frac{1}{2}\rangle$ in eigenstates of $\hat{S}_x$ and finding the probability amplitudes for the two possible outcomes, $m_x = \frac{1}{2}$ and $m_x = -\frac{1}{2}$. Thus

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = a \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + b \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

We find the amplitudes a and b by orthogonal projection in the usual way:

$$a = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}},$$

$$b = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}.$$

We have then for the desired probabilities:

$$\begin{aligned} \text{probability of getting } m_x = \frac{1}{2} & \text{ is } |a|^2 = \frac{1}{2}, \\ \text{probability of getting } m_x = -\frac{1}{2} & \text{ is } |b|^2 = \frac{1}{2}. \end{aligned}$$

Since each particle is therefore equally likely to be found with $m_x = \frac{1}{2}$ or $m_x = -\frac{1}{2}$, equal numbers, on average, go into each of the two emergent beams and so the two beams will have *equal intensity*.

Regeneration

What happens if we select only those particles with $m_x = \frac{1}{2}$ emerging from the second magnet and allow them to impinge on a third magnet whose mean field is aligned with the z -direction? This is the situation illustrated in Fig. 9.2. We are, in effect, remeasuring $\hat{S}_z$ by means of the third apparatus. We know that the state of particles entering the third magnet is $|s = \frac{1}{2}, m_x = \frac{1}{2}\rangle$, and we can expand this state in terms of the complete set of eigenstates of $\hat{S}_z$. The expansion coefficients will be the probability amplitudes required to compute the probabilities of getting the two possible outcomes $m = \frac{1}{2}$ and $m = -\frac{1}{2}$ when we measure $\hat{S}_z$ for each particle:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

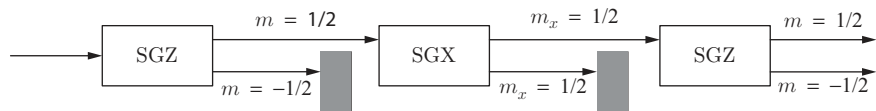


Fig. 9.2

A beam of atoms, each with $s = \frac{1}{2}$, travelling along the y -axis passes through a sequence of Stern–Gerlach magnets whose mean fields are oriented along either the z -direction (SGZ) or the x -direction (SGX). The shaded boxes represent absorbers.

We see that the desired amplitudes are both $\frac{1}{\sqrt{2}}$, so giving equal probabilities for the two outcomes. The remarkable feature of this result is that the probability of getting $m = -\frac{1}{2}$ is non-zero *despite our having eliminated the beam with $m = -\frac{1}{2}$ which emerged from the first magnet!* This phenomenon is referred to as *regeneration*. It has arisen here because the second measurement, of the x -component of angular momentum, was *incompatible with the first measurement*, of the z -component.

General Remarks

More generally, if the second apparatus is aligned so that its mean field lies not in the x -direction, but in the x - z plane at an angle θ to the z -axis, then it measures the component of angular momentum not along the x -direction but along the direction of a unit vector

$$\underline{n} = \sin \theta \underline{e}_x + \cos \theta \underline{e}_z,$$

where $\underline{e}_x$ and $\underline{e}_z$ are the usual Cartesian unit vectors in the x - and z -directions, respectively. The relevant eigenstates are then those of the matrix

$$\underline{\sigma} \cdot \underline{n} = \sigma_x \sin \theta + \sigma_z \cos \theta = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}.$$

We shall investigate some of the properties of spin along a generic direction $\underline{n}$ in the problems at the end of the chapter.

We will come back to discussing Stern–Gerlach in more detail when we introduce entanglement later in this book.

Summary

- The Stern–Gerlach experiment. Experimental evidence for half-integer spin in nature. (9.1)
- We describe the Hilbert space of physical states with spin $\frac{1}{2}$. After choosing a complete set of commuting observables and the corresponding basis, we can expand arbitrary states in that basis and discuss the physical interpretation of the coefficients in the expansion. Spin and orbital properties of a quantum system. (9.2)
- Following what we did in the case of the orbital angular momentum, we can construct a matrix representation of the angular momentum operators for $s = \frac{1}{2}$. (9.3)
- We study the eigenvectors and eigenvalues of the spin component in arbitrary directions. We look at explicit examples in order to get familiar with the Hilbert space of physical states of a system with spin $\frac{1}{2}$. (9.4–9.6)
- Having developed an understanding of the states that are associated with the spin degrees of freedom, we can revisit the Stern–Gerlach experiment and look at some peculiar physical properties. (9.7)

Problems

- 9.1** The Hamiltonian that describes the interaction of a static spin- $\frac{1}{2}$ particle with an external magnetic field, $\underline{B}$, is

$$\hat{H} = -\hat{\underline{\mu}} \cdot \underline{B},$$

where the magnetic moment operator, $\hat{\underline{\mu}}$, is related to the spin operator, $\hat{\underline{S}}$, by $\hat{\underline{\mu}} = \gamma \hat{\underline{S}}$, with γ the gyromagnetic ratio. Use the Pauli representation of the spin vector $\hat{\underline{S}}$ to find the energy eigenvalues in a static uniform magnetic field in the z -direction, $\underline{B}_0 = (0, 0, B_0)$.

- 9.2** Construct the matrix $\underline{\sigma} \cdot \underline{e}$, where $\underline{e}$ is a unit vector with Cartesian components e_x, e_y, e_z . $\underline{\sigma}$ has Cartesian components which are the Pauli matrices, $\sigma_x, \sigma_y, \sigma_z$, so that

$$\underline{\sigma} \cdot \underline{e} = e_x \sigma_x + e_y \sigma_y + e_z \sigma_z.$$

Show that the eigenvalues of $\underline{\sigma} \cdot \underline{e}$ are ± 1 and hence deduce that a measurement of the component of spin along the direction of $\underline{e}$, of a spin- $\frac{1}{2}$ particle, can only yield the result $\frac{1}{2}\hbar$ or $-\frac{1}{2}\hbar$. Obtain the *normalised* eigenvectors of $\underline{\sigma} \cdot \underline{e}$ corresponding to each of the eigenvalues and verify your answers by considering the special case $e_x = 1, e_y = e_z = 0$. Remember that the normalisation condition for a two-component column matrix with entries ψ_1 and ψ_2 is

$$|\psi_1|^2 + |\psi_2|^2 = 1.$$

- 9.3** A beam of spin- $\frac{1}{2}$ particles is sent through a Stern–Gerlach apparatus which divides the incident beam into two spatially separated beams having $m = \pm \frac{1}{2}$, respectively. The beam with $m = -\frac{1}{2}$ is removed, whilst the beam with $m = \frac{1}{2}$ is allowed to impinge on a second Stern–Gerlach apparatus whose mean field is also perpendicular to the beam direction, but inclined at an angle θ with respect to that of the first apparatus. Calculate the relative intensities of the two emergent beams.
- 9.4** Construct the four-dimensional representation of the components of the intrinsic angular momentum $\hat{S}_x, \hat{S}_y, \hat{S}_z$ acting in the space of states with spin $s = 3/2$. Write explicitly the matrix corresponding to the operator $\hat{S}^2$; verify that you obtain the expected result.
- 9.5** The state of a particle of spin $\frac{1}{2}$ in one dimension is described by a two-component wave function in the basis of eigenstates of $\hat{S}_z$:

$$\Psi(x) = \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix}.$$

Let us consider a Hamiltonian with a potential that depends on the spin of the particle:

$$\hat{H} = \frac{1}{2m} \left[\hat{p}^2 + W(\hat{X})^2 + 2\hat{S}_z W'(\hat{X}) \right],$$

where $\hat{P} = -i\hbar \frac{d}{dx}$ is the momentum operator, $\hat{X}$ is the position operator, $\hat{S}_z$ is the operator associated with the z -component of the spin and $W(x)$ is a real function of the position x , such that $|W| \rightarrow \infty$ for $x \rightarrow \pm\infty$ and $W'(x) = \frac{d}{dx}W(x)$. The spin operator is $\hat{S}_i = \frac{\hbar}{2}\sigma_i$, where σ_i are the Pauli matrices:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Compute the action of each term in the Hamiltonian on the wave function specified above. In particular, show that

$$2\hat{S}_z W'(\hat{X})\Psi(x) = \begin{pmatrix} \hbar W'(x)\psi_1(x) \\ -\hbar W'(x)\psi_2(x) \end{pmatrix}.$$

Show that

$$\hat{Q}_1^2 = \hat{Q}_2^2 = \hat{H},$$

where

$$\hat{Q}_1 = \frac{1}{\sqrt{2m}} [\sigma_1 \hat{P} + \sigma_2 W(\hat{X})], \quad \hat{Q}_2 = \frac{1}{\sqrt{2m}} [\sigma_2 \hat{P} - \sigma_1 W(\hat{X})]$$

and σ_1, σ_2 are the Pauli matrices.

Prove that for any state $|\Psi\rangle$:

$$\langle \Psi | \hat{H} | \Psi \rangle \geq 0.$$

Show that the energy of the ground state vanishes if its wave function satisfies

$$[\hat{P} + iW(\hat{X})\sigma_3] \Psi_0(x) = 0.$$

Are $\hat{H}$ and $\hat{S}_z$ compatible observables?

Let us denote by $\Psi^{(n,\pm)}(x)$ the simultaneous eigenstates of $\hat{H}$ and $\hat{S}_z$, with respective eigenvalues E_n and $\pm\hbar/2$. Show that

$$\Psi^{(n,+)}(x) = \begin{pmatrix} \psi_1^{(n,+)}(x) \\ 0 \end{pmatrix}.$$

What conditions need to be satisfied for the ground-state wave function $\Psi^{(0,+)}$ with $E_0 = 0$ to be normalisable?

Prove that $[\hat{Q}_1, \hat{H}] = 0$.

What can you deduce about the degeneracy of the energy eigenvalues?